Toward exact canonical good-Boussinesq scattering
Poisson contraction, soliton sectors and current deformations

Quantum Boussinesq solitons project
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4 November 2026

Abstract

(c) A coordinated investigation tests the proposed quantum-model route beyond the equation-level dictionary. The standard W_3 construction quantises the second Gelfand–Dickey bracket, whereas the specified potential theory uses the first. A singular large-spin-background contraction recovers the canonical bracket with an explicit $\hbar$ normalization. Its multiplicative Baxter arguments acquire finite additive shifts whose spacing in the soliton variable ℓ is $9\hbar/2$. The inherited highest-weight annihilation conditions are incompatible with a positive norm and the required real target fields. The normalization and existence of the limiting Baxter operators remain open.

(c) Independently, an anchored current deformation gives an exact classical Hamiltonian family. It preserves isolated-soliton energy and momentum and has first-order two-body action response $-g(P_1M_2-P_2M_1)$. Its leading local density identifies a dynamical matching parameter that one-body energy and vacuum elasticity miss. A Gaussian infrared calculation shows that changing soliton charges requires a nontrivial state dictionary even with arbitrarily cheap finite-volume compensation.

(d) The exact physical S-matrix and the seed phase coefficient remain undetermined. The constructive outcome is an explicit candidate algebraic contraction, a relative scattering deformation, and quantitative tests for the physical representation and soliton states needed to complete the construction.

1 Scope and research verdict

Claims use the project's meanings of (a), literature result, (b), literature heuristic, (c), project derivation or computation, and (d), unestablished identification or research proposal. Conditional algebraic and Gaussian results retain their stated scope. Every new displayed (c) formula is checked by the script named in its subsection. The four scripts live in `phase2/scripts/` and write only their adjacent JSON caches.

(c) The work followed the proposed canonical real-form and one-soliton dictionary test, an independent conserved-current deformation calculation, and an independent sector audit. The two latter investigations cross-reviewed the contraction, normalization and domain qualifications. A further first-bracket conservation-law diagnostic was derived separately. Existing packages, controlling audits and the previous `shiftcharge_oneloop.tex` were used as context. The optional `audit_good_renorm.tex` and `audit_defect_bootstrap.tex` were absent from the repository inventory. No old script was run or imported.

(c) The fixed target is

$$H_0 = \int \left[\frac{\pi^2 + u^2 + u_x^2}{2} - u^3 \right] \mathrm{d}x, \qquad u = \phi_x, \quad \{\phi(x), \pi(y)\} = \delta(x-y). \tag{1}$$

It gives $u_t = \pi_x$ and $\pi_t = (u - 3u^2 - u_{xx})_x$. Its charges are $M = \int u\,\mathrm{d}x$, $\Pi = \int \pi\,\mathrm{d}x$, and $P = -\int \pi u\,\mathrm{d}x$. The finite quadratic subtraction $\lambda\hbar \int u^2 \mathrm{d}x$ remains a parameter. **(d)** Transferring

source normal ordering to a physical target representation is required before any value of λ can be inferred.

Question	Result established	Remaining input
Direct W_3 identification	**(c)** Brackets disagree at the target vacuum	**(d)** A change of representation or a singular limit
Canonical contraction	**(c)** Correct first bracket, energy and $\hbar$ scale	**(d)** Positive representation and operator convergence
Baxter equation	**(c)** Finite argument shifts and the $9\hbar/2$ unit	**(d)** Vacuum-normalized functions and soliton eigenstates
$P \wedge M$ freedom	**(c)** Explicit classical current-deformation direction	**(d)** Absolute seed scattering and quantum generator
Soliton state space	**(c)** Infrared norm and finite-volume compensation tests	**(d)** Interacting dressing and scattering limits
Exact physical S-matrix	**(d)** Undetermined	**(d)** The preceding physical inputs and channel closure

2 The canonical dictionary and its obstruction

2.1 Source bracket and independent reconstruction

(a) Bazhanov–Hibberd–Khoroshkin (BHK) explicitly identify their classical algebra as the second Gelfand–Dickey structure. Their scalar Lax operator is $D^3 - UD + V - U_x/2 - \lambda_{\mathrm{sp}}^3$. Hikami supplies a lattice chiral algebra and a third-order Baxter relation. The relevant full-text passages were read as recorded in Section 9. [2, 1]

(c) (`shiftcharge_bridge_hikami.py`) With coefficients placed to the left of $D = \partial_x$, the second bracket is independently reconstructed as

$$(J_2)_{UU} = 2D^3 - 2UD - U_x, \qquad (J_2)_{UV} = -3VD - 2V_x,$$
$$(J_2)_{VU} = -3VD - V_x.$$

$$(J_2)_{VV} = -\frac{D^5}{6} + \frac{5U}{6}D^3 + \frac{5U_x}{4}D^2 + \left(\frac{3U_{xx}}{4} - \frac{2U^2}{3}\right)D + \frac{U_{xxx}}{6} - \frac{2UU_x}{3}. \tag{2}$$

The calculation starts from the source Miura fields

$$U = a^2 + b^2 + 2a_x, \qquad \sqrt{3}V = \frac{2}{3}b^3 - 2a^2b - 3ab_x - a_xb - b_{xx}, \tag{3}$$

and the current bracket $-\frac{1}{2}D$ in each component. Composing their Fréchet derivatives with this bracket and the formal adjoint reproduces all four blocks of (2). Formal skew-adjointness is checked separately.

(c) The exact real PDE dictionary is

$$U = \frac{1}{4} - \frac{3}{2}u, \qquad V = \frac{\sqrt{3}}{4}\pi, \qquad \tau = \sqrt{3}\,t. \tag{4}$$

The source flow $U_\tau = -2V_x$, $V_\tau = U_{xxx}/6 - 2UU_x/3$ then gives (1). However the target canonical bracket in these variables is

$$J_1 = -\frac{3\sqrt{3}}{8}\begin{pmatrix} 0 & D \\ D & 0 \end{pmatrix}. \tag{5}$$

Its UU bracket vanishes, whereas (2) has a third derivative and field-dependent terms. Scalar rescaling of time or $\hbar$ cannot identify the displayed observable maps.

2.2 How far the obstruction extends

(c) (`shiftcharge_bridge_hikami.py`) At $U = 1/4, V = 0$,

$$J_2 = \operatorname{diag}\left(2D(D^2 - \tfrac{1}{4}), -\frac{D(D^2-1)(D^2-\frac{1}{4})}{6}\right), \qquad \frac{\det J_2}{\det J_1} = \frac{64}{81}(D^2 - \tfrac{1}{4})^2(D^2 - 1). \tag{6}$$

The nonconstant determinant ratio excludes a finite-order differential linear change with a finite-order differential inverse at this vacuum. More generally, for constant backgrounds the Hermitian Fourier symbol is

$$iJ_2(iq) = q\begin{pmatrix} 2(q^2+U) & 3V \\ 3V & (q^2+U)(q^2+4U)/6 \end{pmatrix}, \tag{7}$$

with determinant

$$q^2\left[\frac{(q^2+U)^2(q^2+4U)}{3} - 9V^2\right]. \tag{8}$$

At the stated vacuum and $q > 0$, $iJ_2(iq)$ is positive definite and $iJ_1(iq)$ has one eigenvalue of each sign. Inertia under Hermitian congruence excludes a real invertible translation-invariant linear Fourier map at this same vacuum, including a nonlocal one. **(d)** Changes of vacuum, complex reality conditions and singular limits require separate tests.

3 A singular contraction with the correct quantum scale

3.1 Poisson pencil and Hamiltonian

(c) (`shiftcharge_bridge_hikami.py`) A concrete escape from the fixed-vacuum obstruction is

$$V_{\text{source}} = V + \Lambda, \qquad \kappa_\Lambda = \frac{\sqrt{3}}{8\Lambda}, \qquad \lim_{\Lambda\to+\infty} \kappa_\Lambda J_2(U, V+\Lambda) = J_1. \tag{9}$$

The limit holds coefficient by coefficient for fields of finite size. The shifted bracket has a large off-diagonal first-derivative term, which survives this rescaling.

(c) Let $F(U,V) = U^3 + 9V^2 + 3U_x^2/4$. The physical Hamiltonian has the exact density representation

$$H_0 = \frac{8}{27}\int\left[F(U,V) - \frac{3}{16}U + \frac{1}{32}\right]\mathrm{d}x. \tag{10}$$

Direct substitution of (4) verifies this identity, and J_1 acting on its functional gradient gives physical t evolution. The required large-background subtraction obeys

$$F(U, V+\Lambda) - 18\Lambda(V+\Lambda) + 9\Lambda^2 = F(U,V). \tag{11}$$

Thus a shifted higher integral with lower-integral subtractions supplies a candidate Hamiltonian. The source's Hamiltonian time and its normalized circle integrals must be converted explicitly.

(a) BHK's spin-six quantum density contains $:T^3: + 9 :W^2: + (c_{\mathrm{V}} - 10) :T_x^2: /32$. **(c)** With $T = (c_{\mathrm{V}}/24)U$ and $W = i(-c_{\mathrm{V}}/24)^{3/2}(V+\Lambda)$, division by $(c_{\mathrm{V}}/24)^3$ gives the derivative coefficient

$$\frac{3}{4}\left(1 - \frac{10}{c_{\mathrm{V}}}\right). \tag{12}$$

(d) The source regularized product is part of this algebraic prescription. Its limit in a new physical representation, including the finite subtractions, has yet to be constructed.

3.2 Exact mode normalization and failure of the inherited vacuum

(c) (`shiftcharge_bridge_hikami.py`) On a circle of circumference 2π, use the source e^{-inx} convention and define

$$U_n = \frac{24}{c_{\mathrm{V}}}\left(L_n - \frac{c_{\mathrm{V}}}{24}\delta_{n0}\right), \qquad V_n = \frac{W_n}{i(-c_{\mathrm{V}}/24)^{3/2}} - \Lambda\delta_{n0}. \tag{13}$$

The exact mixed source commutator gives

$$[U_n, V_m] = \frac{24}{c_{\mathrm{V}}}(2n-m)\left(V_{n+m} + \Lambda\delta_{n+m,0}\right). \tag{14}$$

Choosing

$$c_{\mathrm{V}}(\Lambda) = -\frac{128\sqrt{3}\pi\Lambda}{\hbar} \tag{15}$$

makes its central term precisely $-3\sqrt{3}\hbar n\,\delta_{n+m,0}/(16\pi)$, the Fourier form of (5). The UU and VV commutators vanish in the coefficientwise contraction at fixed mode indices and finite rescaled fields. This establishes an algebraic $\hbar$ dictionary.

(c) The required physical involution is $U_n^\dagger = U_{-n}$, $V_n^\dagger = V_{-n}$. With the factor of i in (13), this requires $L_n^\dagger = L_{-n}$ and $W_n^\dagger = -W_{-n}$ when $c_{\mathrm{V}} < 0$. Retaining the source highest-weight annihilation conditions $U_n|0\rangle = V_n|0\rangle = 0$ for $n > 0$ then gives the limiting Gram matrix

$$G_n \longrightarrow \begin{pmatrix} 0 & K_n \\ K_n & 0 \end{pmatrix}, \qquad K_n = -\frac{3\sqrt{3}\hbar n}{16\pi}. \tag{16}$$

Its eigenvalues have opposite signs. Already the exact UU entry is $48n(n^2+U_0)/c_{\mathrm{V}}$, which is negative at $n = 1, U_0 = 1/4$. At $\Lambda = 1000, \hbar = 1$ the two independently evaluated level-one eigenvalues are approximately -0.1034244254 and 0.1033239228. This excludes the inherited annihilation conditions with the required target reality structure as a positive vacuum.

(c) For constant real Miura currents, (3) also implies

$$4U^3 - 27V^2 = 4a^2(a^2-3b^2)^2 \geq 0. \tag{17}$$

At $U = 1/4$ it requires $|V| \leq 1/(12\sqrt{3})$. The large-Λ path therefore leaves the real constant-Miura sector. **(d)** A different polarization or a different representation could still realise the contracted canonical algebra. The existing CFT eigenstates do not yet supply that construction.

3.3 Baxter arguments become additive

(a) BHK identify the transfer-matrix spectral parameter as $t_{\mathrm{sp}} = \lambda_{\mathrm{sp}}^3$, and give the exact relation

$$Q(t_{\mathrm{sp}}q^3) - T(t_{\mathrm{sp}}q^{1/2})Q(t_{\mathrm{sp}}q) + \overline{T}(t_{\mathrm{sp}}q^{-1/2})Q(t_{\mathrm{sp}}q^{-1}) - Q(t_{\mathrm{sp}}q^{-3}) = 0. \tag{18}$$

Here $q = e^{i\pi\mathfrak{g}}$ and $c_{\mathrm{V}} = 50 - 24(\mathfrak{g} + \mathfrak{g}^{-1})$. The bar labels the second transfer operator.

(c) (`shiftcharge_bridge_hikami.py`) Select the small positive $\mathfrak{g}_\Lambda$ root and put $t_{\mathrm{sp}} = \Lambda + z$. Equations (15) and (18) give

$$\eta = \frac{\sqrt{3}\hbar}{16}, \qquad A_r(z) = q_\Lambda^r(\Lambda + z) - \Lambda \longrightarrow z + ir\eta. \tag{19}$$

For $Q_\Lambda(z) = Q(\Lambda + z)$, and analogously for the transfer operators, the finite-Λ relation is exactly

$$\begin{aligned} &Q_\Lambda(A_3(z)) - T_\Lambda(A_{1/2}(z))Q_\Lambda(A_1(z)) \\ &\qquad + \overline{T}_\Lambda(A_{-1/2}(z))Q_\Lambda(A_{-1}(z)) - Q_\Lambda(A_{-3}(z)) = 0. \end{aligned} \tag{20}$$

The argument limits are checked symbolically and with 80-digit arithmetic. At $\Lambda = 10^{20}, \hbar = 1, z = 0.70$ the largest of the six sampled argument errors is below 2.2×10^{-21}.

(d) If compatible normalizations and domains give limits of the operators or eigenfunctions, their prospective equation is

$$Q(z+3i\eta) - T(z+i\eta/2)Q(z+i\eta) + \overline{T}(z-i\eta/2)Q(z-i\eta) - Q(z-3i\eta) = 0. \tag{21}$$

A z-dependent scalar renormalization of Q_Λ changes the coefficients. Establishing these normalized limits is a dynamical step.

(c) The soliton roots

$$\mu_\pm = \frac{-v/\sqrt{3} \pm k}{2}, \qquad k^2 + v^2 = 1, \qquad \ell = v(4k^2 - 1) \tag{22}$$

satisfy

$$\mu_+^3 - \frac{\mu_+}{4} = \mu_-^3 - \frac{\mu_-}{4} = z = -\frac{\ell}{12\sqrt{3}}. \tag{23}$$

Successive Q arguments in (21) are separated by $2i\eta$, giving

$$\Delta\ell = -12\sqrt{3}(2i\eta) = -\frac{9}{2}i\hbar. \tag{24}$$

This agrees with the unit in the project's semiclassical string condition. **(d)** Identification of actual roots, physical poles or fusion amplitudes requires the limiting eigenfunctions and their state interpretation.

4 A dynamical realization of the antisymmetric direction

4.1 Currents and the anchored generator

(a) Generalized current deformations produce relative scattering dressings under specified locality and scattering assumptions. The source treatments retain finite-volume boundary terms and allow momentum-dependent charge eigenvalues. Their full-text derivations motivate the following independent Boussinesq calculation. [4, 5]

(c) (`shiftcharge_bridge_deformation.py`) The target conservation laws have densities and currents

$$q_M = u, \quad j_M = -\pi, \qquad q_P = p = -\pi u, \quad j_P = \frac{\pi^2 + u^2 + u_x^2}{2} - 2u^3 - uu_{xx}. \tag{25}$$

Both continuity equations are checked directly from (1). For a fixed left endpoint ϕ_-, let $\chi = \phi - \phi_-$ and set

$$X = -\int_{x<y} u(x)p(y)\,\mathrm{d}x\mathrm{d}y = \int \chi\pi u\,\mathrm{d}x. \tag{26}$$

The integrated current cross product is

$$\begin{aligned} \mathcal{D} &= q_M j_P - j_M q_P = -\frac{1}{2}\pi^2 u + \frac{1}{2}u^3 - 2u^4 - u^2u_{xx} + \frac{1}{2}uu_x^2, \\ \int \mathcal{D}\,\mathrm{d}x &= \int \mathcal{D}_{\mathrm{bulk}}\,\mathrm{d}x - [u^2u_x], \\ \mathcal{D}_{\mathrm{bulk}} &= -\frac{1}{2}\pi^2 u + \frac{1}{2}u^3 - 2u^4 + \frac{5}{2}uu_x^2. \end{aligned} \tag{27}$$

For localized derivatives and vanishing currents the boundary term vanishes, and $\partial_g H_g = \{H_g, X\}$ at $g = 0$ gives this perturbation. An independently checked local identity is

$$\partial_t(\chi\pi u) = \partial_x(\chi j_P) - \mathcal{D}. \tag{28}$$

4.2 The exact classical family

(c) (`shiftcharge_bridge_deformation.py`) The anchored canonical flow is

$$\partial_g \chi = \chi \chi_x, \qquad \partial_g \pi = \chi \pi_x. \tag{29}$$

Its characteristic solution is

$$x = y - g\chi_0(y), \quad \chi_g(x) = \chi_0(y), \quad \pi_g(x) = \pi_0(y), \quad u_g(x) = \frac{u_0(y)}{1 - g u_0(y)}. \tag{30}$$

The chart requires $1 - g u_0(y) > 0$. Writing $J = 1 - g u_0$, the variations at fixed x obey $\delta\chi_g = \delta\chi_0/J$ and $\delta\pi_g = \delta\pi_0 + g\pi_0' \delta\chi_0/J$. Since $\mathrm{d}x = J\mathrm{d}y$, their wedge product preserves the canonical symplectic form.

(c) Transforming the Hamiltonian gives the all-order classical expression

$$\boxed{H_g = \int \left[\frac{\pi^2(1-gu)}{2} + \frac{u^2}{2(1-gu)} - \frac{u^3}{(1-gu)^2} + \frac{u_x^2}{2(1-gu)^5} \right] \mathrm{d}x.} \tag{31}$$

Differentiation at $g = 0$ reproduces (27). **(d)** Quantum ordering, contact subtractions and a global self-adjoint generator have yet to be constructed. The possible singularity of (30) is a concrete limitation of the classical chart.

(c) Within that anchored chart the field integrals obey

$$T_g M = M, \qquad T_g P = P, \qquad T_g \Pi = \Pi + gP. \tag{32}$$

Here the left endpoint is a fixed c-number. **(d)** If it is promoted to a dynamical global shift variable, the complete boundary symplectic problem must be solved. A uniform shift of ϕ and ϕ_- together leaves the bilocal expression invariant. The formal constant-shift vector field is also outside the fixed-endpoint tangent space. Consequently (32) is used as a verified anchored field-integral response, without asserting a global quantum commutator.

4.3 One-body invariance and the centroid identity

(c) (`shiftcharge_bridge_deformation.py`) A solution of H_g is obtained by applying T_{-g} to a solution of H_0. For an isolated soliton,

$$x = y + g\chi_0(y,t), \qquad u_{\mathrm{phys}} = \frac{u_0}{1 + g u_0}, \qquad \pi_{\mathrm{phys}} = -v u_0. \tag{33}$$

Direct integration, separately from the canonical argument, gives

$$M_g = M_0, \quad P_g = P_0, \quad E_g = E_0, \quad \Pi_g = \Pi_0 - gP_0, \quad B_g = B_0 + \frac{gM_0^2}{2}, \qquad B = \int xu \, \mathrm{d}x. \tag{34}$$

The centroid retains its original velocity. The deformed Hamiltonian has $\dot{\phi} = (1 - gu)\pi$, so its integrated continuity identity is

$$-M v_{\mathrm{cm}} = \int (1 - gu)\pi \, \mathrm{d}x = \Pi + gP. \tag{35}$$

The original identity uses the original kinetic Hamiltonian. This calculation explains how a proportional shift of Π can accompany an unchanged dispersion in a different Hamiltonian family.

(c) On a classical one-soliton profile, $\pi = -vu$, $v^2 = 1 - k^2$, and $u_x^2 = k^2u^2 - 2u^3$ imply

$$\mathcal{D}_{\mathrm{bulk}} = 3k^2u^3 - 7u^4, \qquad \int u^3 \mathrm{d}x = \frac{4k^5}{15}, \quad \int u^4 \mathrm{d}x = \frac{4k^7}{35}, \quad \int \mathcal{D}_{\mathrm{bulk}} \mathrm{d}x = 0. \tag{36}$$

The vacuum potential in (31) has $V_g''(0) = 1$ exactly. For $g = O(\hbar)$, this direction has no quadratic vacuum term or one-soliton energy shift at order $\hbar$. Its independent cubic kinetic operator $-\pi^2 u/2$ also distinguishes it from λu^2. Thus these one-body and vacuum measurements do not determine this relative scattering deformation at that order.

4.4 Two-body phase and an independent collision test

(c) (`shiftcharge_bridge_deformation.py`) Fix incoming ordering $x_1 < x_2$ and outgoing ordering $x_1 > x_2$. For isolated symmetric solitons, $X_j = -M_j P_j/2$. The separated generator has limits

$$X_{\text{in}} = X_1 + X_2 - M_1 P_2, \qquad X_{\text{out}} = X_1 + X_2 - M_2 P_1. \tag{37}$$

Consequently

$$X_{\text{out}} - X_{\text{in}} = -(P_1 M_2 - P_2 M_1) = -Q_{12}, \qquad \Delta I = -g \int \mathcal{D}\,\mathrm{d}x\mathrm{d}t = -gQ_{12}. \tag{38}$$

The last equality is the first-order fixed-momentum action response. It gives a leading semiclassical relative factor

$$\exp\left[-\frac{ig}{\hbar} Q_{12}\right]. \tag{39}$$

Taking $g = -\hbar c_{\text{CDD}}$ realises the displayed order-one direction. This is the full relative phase response. Computing only the response of the old topological cross term would omit other dynamical and frame terms.

(c) The checker constructs an independent exact two-soliton tau function at $k_1 = 0.25, k_2 = 0.60$, both right-moving. It first tests the canonical field equations, then integrates X at $t = \pm 2500$. It finds

$$\begin{aligned} X_{\text{in}} &= -0.12924147353268712036795525091131 67, \\ X_{\text{out}} &= -0.08374454648958529813414045528563667, \\ X_{\text{out}} - X_{\text{in}} &= 0.04549692704310182223381479562568000 = -Q_{12}. \end{aligned} \tag{40}$$

The incoming and outgoing factorizations each agree within 2×10^{-47} against a declared 10^{-37} tolerance. A separate calculation at the collision gives

$$\int \mathcal{D}_{\text{bulk}}(x, 0)\,\mathrm{d}x = -0.00045398229875116660705088019041 28653, \tag{41}$$

and verifies it against $-\dot{X}$ to better than 10^{-65}. This is a current-insertion check through a collision. The projected functional determinant remains uncomputed.

(d) If a self-adjoint renormalized X, a common domain and asymptotic soliton states exist, the candidate unitary is $U_g = e^{igX/\hbar}$. With $H_g = U_g H_0 U_g^\dagger$ and a fixed asymptotic coordinate identification, the relative quantum relation would be

$$S_g(P_1, P_2) = e^{-igQ_{12}/\hbar} S_0(P_1, P_2). \tag{42}$$

Transforming the observables and asymptotic identification along with the Hamiltonian gives the corresponding equivalent description. Holding the laboratory coordinate, field and circumference fixed makes (31) a different Hamiltonian representative.

(d) Equation (42) leaves the seed S_0 undetermined. Specifying $g = 0$ selects the original member of this family. It does not calculate the coefficient called c in an incomplete seed-phase ansatz. The full functional continuum ambiguity also remains unclassified.

4.5 Finite circumference retains a boundary contribution

(c) (`shiftcharge_bridge_deformation.py`) On an interval $[a, b]$, the anchored generator is

$$X = -\int_a^b p(y) \int_a^y u(x)\mathrm{d}x\mathrm{d}y.$$

Using the primitive anchored at a at each time, the continuity equations give

$$-\dot{X} = \int_a^b \mathcal{D}\mathrm{d}x + j_M(a)P - M j_P(b). \tag{43}$$

At periodic endpoints this reads $\int \mathcal{D} = -\dot{X} + M j_P(a) - j_M(a) P$. A finite-interval numerical profile with nonzero endpoint currents checks the complete identity independently to better than 10^{-65}. The reciprocal coordinate map sends a period L with potential twist M to $L - gM$. **(d)** Identifying $-\dot{X}$ with $\{H, X\}$ requires a boundary phase space in which both functionals have Hamiltonian vector fields.

(c) The corresponding algebraic ring-phase contribution is

$$\sum_l \Delta\delta_{jl} = -\frac{g}{\hbar}(P_j M_{\rm tot} - M_j P_{\rm tot}),$$
$$\frac{P_j L}{\hbar} + \sum_l \Delta\delta_{jl} = \frac{P_j(L - gM_{\rm tot})}{\hbar} + \frac{g M_j P_{\rm tot}}{\hbar}. \tag{44}$$

(d) Deriving a physical finite-volume spectrum requires a fixed circumference and independently defined seam. The length response and the charge-dependent twist must be kept in that dictionary.

5 The soliton sector has a measurable infrared requirement

5.1 Conserved charges and the finite-ring zero mode

(c) (`shiftcharge_bridge_sector.py`) In the derivative-field algebra the first Hamiltonian operator is $\left(\begin{smallmatrix} 0 & D \\ D & 0 \end{smallmatrix}\right)$. Constant gradients show that M and Π are central for compactly supported derivative-field observables. In the potential algebra, $\{\phi(x), \Pi\} = 1$. With decaying bulk fields,

$$\dot{M} = [\pi], \qquad \dot{\Pi} = [u - 3u^2 - u_{xx}]. \tag{45}$$

Endpoint variables need their own phase-space treatment.

(c) The traveling branch has

$$M = 2k, \qquad \Pi = -2kv, \qquad P = \frac{2}{3}k^3 v = \frac{vM^3}{12}, \qquad v = \epsilon\sqrt{1 - M^2/4}. \tag{46}$$

Fixing its kink height fixes the amplitude, leaving discrete velocity signs. This bare classical family supplies no open momentum interval inside one fixed-height sector. **(d)** Dressed field states can carry other momenta in that sector. Their relation to derivatives along the classical $E(P)$ branch has to be specified.

(c) At fixed twist M on a ring,

$$\phi(x) = \Phi + \frac{Mx}{L} + \widetilde{\phi}(x), \qquad \pi(x) = \frac{\Pi}{L} + \widetilde{\pi}(x),$$
$$\Omega = \mathrm{d}\Pi \wedge \mathrm{d}\Phi + \int \mathrm{d}\widetilde{\pi} \wedge \mathrm{d}\widetilde{\phi}. \tag{47}$$

The tilded fields are periodic with zero mean. The uniform configuration has

$$H_{\rm unif} = \frac{\Pi^2 + M^2}{2L} - \frac{M^3}{L^2}, \qquad P_{\rm unif} = -\frac{M\Pi}{L}. \tag{48}$$

(d) Compactification of Φ, allowed values of M, and quantum boundary conditions remain additional data.

5.2 Positive canonical oscillator polarization

(c) (`shiftcharge_bridge_sector.py`) The positive quadratic-vacuum representation can be written explicitly. For nonzero ring modes with $q_n = 2\pi n/L$, take $\phi_n^\dagger = \phi_{-n}$, $\pi_n^\dagger = \pi_{-n}$ and $[\phi_n, \pi_m] = i\hbar\delta_{n+m,0}/L$. Define

$$b_n = \sqrt{\frac{L}{2\hbar}}\left(\sqrt{\omega_n}\phi_n + \frac{i\pi_n}{\sqrt{\omega_n}}\right), \qquad \omega_n = |q_n|\sqrt{1 + q_n^2}. \tag{49}$$

The independent oscillator checks give $[b_n, b_m^\dagger] = \delta_{nm}$, $[b_n, b_m] = 0$, and the inverses

$$\phi_n = \sqrt{\frac{\hbar}{2L\omega_n}}(b_n + b_{-n}^\dagger), \qquad \pi_n = -i\sqrt{\frac{\hbar\omega_n}{2L}}(b_n - b_{-n}^\dagger). \tag{50}$$

At a finite symmetric mode cutoff the quadratic Hamiltonian is

$$H_2 = \sum_{n\neq 0} \hbar\omega_n \left(b_n^\dagger b_n + \frac{1}{2}\right). \tag{51}$$

The modes b_n and b_{-n} are independent annihilation operators. The source conditions that both U_n and V_n annihilate for $n > 0$ would force both b_n and $b_{-n}^\dagger$ to annihilate the state. The latter condition is incompatible with a positive Fock vacuum.

(d) The free canonical polarization is therefore available. The open representation problem is to realise the limiting commuting densities and transfer/Baxter operators with compatible ordering and domains in this polarization, and to extend it to the charged soliton sectors.

5.3 Exact Gaussian overlap test

(c) (`shiftcharge_bridge_sector.py`) Completing the square in the oscillator wavefunctions gives overlap magnitude $e^{-N_{\rm diff}/2}$, where in the good free vacuum

$$N_{\rm diff} = \frac{1}{2\hbar}\int \frac{\mathrm{d}q}{2\pi}\left[\omega(q)|\Delta\widehat{\phi}(q)|^2 + \frac{|\Delta\widehat{\pi}(q)|^2}{\omega(q)}\right], \qquad \omega(q) = |q|\sqrt{1+q^2}. \tag{52}$$

For real initial data the cross term cancels between q and $-q$. At $q \neq 0$, $\Delta\widehat{\phi} = \Delta\widehat{u}/(iq)$. The global additive constant is a separate zero mode. For the exponentially localized soliton profiles considered here, $\Delta\widehat{u}(0) = \Delta M$ and $\Delta\widehat{\pi}(0) = \Delta\Pi$. Hence

$$\boxed{N_{\rm diff}(\mu) = \frac{(\Delta M)^2 + (\Delta\Pi)^2}{2\pi\hbar}\log\frac{1}{\mu} + O(1).} \tag{53}$$

The regulated overlap magnitude has power $[(\Delta M)^2 + (\Delta\Pi)^2]/(4\pi\hbar)$. Changing these charges is therefore an infrared-divergent coherent displacement of the quadratic-vacuum representation on the line. **(d)** This calculation does not determine an interacting infrared exponent or establish the existence or failure of a dressed scattering operator.

(c) The independent Fourier quadrature uses

$$u_k(x) = \frac{k^2}{2}\operatorname{sech}^2(kx/2), \qquad \widehat{u}_k(q) = \frac{2\pi q}{\sinh(\pi q/k)}, \qquad \widehat{\pi}_k(q) = -v\widehat{u}_k(q). \tag{54}$$

For right movers $k_1 = 0.31, k_2 = 0.53$ at $\hbar = 1$, the predicted occupation logarithm coefficient is

$$A_N = 0.046050090668759336609346583574639 06. \tag{55}$$

Independent cutoff integrals at $\mu = 10^{-6}, 10^{-12}, 10^{-24}$ converge after subtracting $A_N \log(1/\mu)$ and reproduce its coefficient to better than 2.0×10^{-26}. A fixed-amplitude translation by 1.7 at $k = 0.31$ instead has the finite limiting occupation 0.00585916183977936915005365274668563 3.

5.4 Compensation in one finite-volume space

(c) (`shiftcharge_bridge_sector.py`) Periodize each localized profile and subtract its own uniform M/L and Π/L pedestals. Both profiles now have the same total charges $M = \Pi = 0$. Their nonzero Fourier coefficients retain (54). For $q_n = 2\pi n/L$,

$$N_L = \frac{1}{\hbar L}\sum_{n\geq 1}\left[\frac{\sqrt{1+q_n^2}}{q_n}|\Delta\widehat{u}(q_n)|^2 + \frac{|\Delta\widehat{\pi}(q_n)|^2}{q_n\sqrt{1+q_n^2}}\right]. \tag{56}$$

These profiles define periodic initial data. The exact nonlinear periodic traveling solution is a separate problem. The independently summed doubling slopes approach A_N.

L	$[N_{2L} - N_L]/\log 2$	Error from A_N
256	0.0461099132406	5.98226×10^{-5}
512	0.0460650331524	1.49425×10^{-5}
1024	0.0460538254672	3.73480×10^{-6}

(c) No fitted logarithm coefficient enters this test. The occupation quadratures and Fourier sums use $q \leq 8$, where the exact hyperbolic-sine factors suppress the tail. The many digits retained in JSON are arithmetic output. The quoted convergence tests, rather than all those digits, are the numerical accuracy claims. The logarithm survives charge compensation in a common finite-volume oscillator space.

(c) A separate construction places a smooth cloud far from the pulse,

$$u_c(x) = \frac{\Delta M}{L} w((x-R)/L), \quad \pi_c(x) = \frac{\Delta \Pi}{L} w((x-R)/L), \quad \int w = 1. \tag{57}$$

It has fixed compensating charges and

$$\begin{aligned} H_c &= \frac{(\Delta M)^2 + (\Delta \Pi)^2}{2L} I_2 - \frac{(\Delta M)^3}{L^2} I_3 + \frac{(\Delta M)^2}{2L^3} I_d, \\ P_c &= -\frac{\Delta M \Delta \Pi}{L} I_2, \end{aligned} \tag{58}$$

where $I_2 = \int w^2$, $I_3 = \int w^3$, and $I_d = \int (w')^2$. For $w = e^{-y^2}/\sqrt{\pi}$ the moments are $I_2 = I_d = 1/\sqrt{2\pi}$ and $I_3 = 1/(\sqrt{3}\pi)$. Direct spatial quadrature checks the complete charges and energy at two widths. Thus vanishing compensation energy and a controlled state normalization are separate requirements.

(d) The quantum dictionary should identify operators intertwining boundary-charge representations or a finite-volume family with controlled soft compensation. The zero mode, nonzero phonon modes, and seam normalization need explicit treatment before extracting a scattering phase.

6 A first-bracket quantum-integrability diagnostic

(c) (`shiftcharge_bridge_hierarchy.py`) An additional conserved density can be derived directly in canonical fields, without using the W_3 representation. Let

$$H_s = \int \left[\frac{\pi^2 + s u^2 + u_x^2}{2} - u^3 \right] \mathrm{d}x. \tag{59}$$

Solving a weighted polynomial ansatz by setting both Euler derivatives of its time derivative to zero gives, modulo a multiple of H_s,

$$\begin{aligned} \mathcal{Q}_7 &= \int \left[\pi_x^2 + u_{xx}^2 - 3u\pi^2 + 3u^4 - 9u u_x^2 + s(u_x^2 - u^3) \right] \mathrm{d}x, \\ \{H_s, \mathcal{Q}_7\} &= 0. \end{aligned} \tag{60}$$

This holds with appropriate vanishing boundary flux or periodic fields. The script solves for the coefficients independently, then verifies the final conservation law. Direct soliton integration at $s = 1$ gives

$$\mathcal{Q}_7[u_k, -v u_k] = \frac{4k^5(20k^2 - 21)}{105}, \tag{61}$$

with independent 70-digit quadrature at three amplitudes.

(c) A finite-dimensional Weyl-symbol degree observation applies to linear canonical regulators. H_s is at most cubic in the coordinate variables and quadratic in momenta, with no mixed kinetic

terms. The charge in (60) is at most quadratic in momenta. The only potentially nonzero third derivative of H_s is purely in coordinates, whereas the corresponding third momentum derivative of $\mathcal{Q}_7$ vanishes. All higher odd Moyal contractions vanish by degree. Thus for these regulated polynomial symbols

$$[\mathrm{Op_W}(H_s), \mathrm{Op_W}(\mathcal{Q}_7)] = i\hbar\,\mathrm{Op_W}(\{H_s, \mathcal{Q}_7\}_{\mathrm{regulator}}). \tag{62}$$

(c) The regulator's classical bracket need not vanish. For the naive periodic three-site central-difference discretization, at $\phi = (1, 2, 4)$ and $\pi = (2, -1, 3)$ with $s = 1$, it is exactly $-75/8$. This rejected regulator is retained as an explicit failure. **(d)** Constructing renormalized continuum operators still requires a compatible regulator, domain and ordering prescription. The finite Weyl identity supplies a diagnostic for that work. Equation (60) also persists for arbitrary s, so this conservation-law test alone cannot select the finite elastic subtraction.

7 What can now be attempted toward the exact amplitude

(d) The first proposed assignment has produced a precise obstruction and a candidate replacement. The remaining programme is organized by observable tests rather than by an assumed scalar-factor ansatz.

1. Realise the limiting commuting densities in the explicit positive canonical polarization (49), replacing the inherited highest-weight annihilation conditions. Determine the finite normal-ordering terms and common domains. Equations (5), (10), (15) and (60) provide explicit checks.

2. Obtain vacuum-normalized limits of the transfer and Baxter operators at $t_{\mathrm{sp}} = \Lambda + z$. Establish convergence or identify an obstruction. The six finite shifts in (19) and the spectral relation (23) fix the proposed canonical scale.

3. Identify one-soliton states or intertwiners with the correct M, Π, P and classical profile. Test their infrared normalization against (53) and (56), keeping the interacting nature of the final exponent as a separate question.

4. Derive two-excitation finite-volume equations in that same representation. Separate the dispersion, soft-state normalization, potential seam and circumference response using (43) and (44). Then extract the phase and compare several kinematic points with a boundary-compatible continuum calculation.

5. Match the coefficient of (27) in fixed canonical fields and physical coordinates. It is the explicit local direction associated with the relative $P \wedge M$ factor. This matching supplements the finite quadratic subtraction.

(c) The resonant-channel audit already requires care with heavy classical solitons. **(d)** The light branch supplies the initial candidate sector. Closure under quantum scattering, including phonon and number-changing channels, must follow from the constructed states and conserved operators.

(d) At this stage there is no extracted physical S_0, no physical two-soliton determinant, and no values of $c_{\mathrm{good}}(\lambda)$ or c_{bad}. The new result is a relative deformation law and a canonical functional-equation route with its representation obstruction identified. The earlier Toda comparison remains method validation with its audited dictionary qualifications.

8 Verification coverage and failures recorded

Claim group	Independent verification	Script
Second versus first bracket	Miura Fréchet reconstruction, formal adjoints, Fourier inertia and discriminant	`shiftcharge_bridge_hikami.py`
Canonical contraction	PDE, energy, bracket coefficients and exact mixed modes	`shiftcharge_bridge_hikami.py`
Representation and Baxter scale	Exact finite-parameter Gram matrices, 80-digit argument limits and spectral polynomial reduction	`shiftcharge_bridge_hikami.py`
Current deformation	Continuity laws, reciprocal Jacobian, symplectic form, exact classical Hamiltonian and one-body integrals	`shiftcharge_bridge_deformation.py`
Scattering and ring response	Exact two-soliton field equations, separated and collision quadratures, finite-interval boundary currents	`shiftcharge_bridge_deformation.py`
Soliton sectors	Positive oscillator algebra, Gaussian integral, exact Fourier transform, logarithm coefficient, independent finite-ring sums and cloud quadratures	`shiftcharge_bridge_sector.py`
Canonical extra charge	Independent coefficient solve, Euler derivatives, soliton quadrature and rejected lattice regulator	`shiftcharge_bridge_hierarchy.py`

(c) The symbolic checks use SymPy. Numerical checks use 70 or 80 decimal digits. The four scripts together run in about one minute on the task machine, with the two-soliton deformation quadrature taking most of the time. The deformation checker records 49 symbolic and numerical checks, including independent charge quadratures and exact-tau collision tests. Each script ends by printing `ALL CHECKS PASSED`. JSON files distinguish computed quantities from unconstructed physical observables. Numerical agreement certifies the stated tests.

1. **(c)** Direct same-vacuum identification fails the UU bracket test. The stronger real Fourier-map attempt fails the inertia test.

2. **(c)** The large-background contraction leaves the real constant-Miura region. The inherited highest-weight annihilation conditions fail positivity under the required target involution.

3. **(c)** The bare traveling-soliton family has no open momentum branch at fixed kink height. Low-energy charge compensation retains an infrared logarithm in its Gaussian state normalization.

4. **(c)** One-body dispersion and vacuum elasticity miss the current deformation at order $\hbar$. This records failure of those observables to fix the relative two-body direction.

5. **(c)** The naive three-site canonical regulator fails the extra-charge commutation test, with exact bracket $-75/8$.

6. **(c)** Promoting the anchored charge response to a global quantum commutator requires additional boundary data. The report retains that qualification explicitly.

7. **(c)** The Belov–Chaltikian HTML retrieval failed. The local full PDF supplied the required passages. BHK's printed equation (2.24) repeats I_4 for the spin-six density, and the cubic printed after equation (2.29) disagrees with its displayed Lax operator. The source pages were visually checked. All spectral calculations here use equation (2.26) and direct polynomial substitution.

8. **(c)** No Python implementation assertion failed and no numerical tolerance was weakened. Two document-authoring submissions initially failed because LaTeX dollar-brace syntax was interpreted by the JavaScript template wrapper. Rephrasing that math command fixed the wrapper before the affected text was written. The first PDF build exposed two overfull lines, repaired by splitting the operator display and the interval formula. A draft numerical-sample label was corrected from $z = 0.37$ to the actual checker input $z = 0.70$ during independent review. The computed bound was unchanged. Review also restricted the logarithm's bounded-remainder statement to the exponentially localized profiles actually studied and stated the interval identity in terms of $-\dot{X}$ before any boundary Poisson identification.

9 Sources and reading depth

(a) Relevant full-text passages were read for each external reference below. The reading did not reproduce every proof in the cited papers.

- Hikami [1], sections 1–2, the Baxter result and analytic assumption in section 3, sections 4–5, conclusion and appendices A–B. Its inversion identity labelled unitarity needs a specified adjoint before being used as physical Hilbert-space unitarity.
- BHK [2], introduction, sections 2.1–2.2 and 3.2–3.3, the spectral convention in section 4.2 and equation (1.11). Printed pages 8–10 were visually inspected. The current work independently checks the target normalization and the required reality structure.
- Belov–Chaltikian [3], sections 1–5, including the lattice brackets, classical bi-Hamiltonian construction and continuum discussion.
- Doyon–Durnin–Yoshimura [4], sections 2–3, appendix B and the boundary formulas in appendix C.
- Cardy–Doyon [5], sections 2.2, 3.2–3.3 and 3.6.

(c) Internal inputs include the README Status, the ranked programme in `report.tex`, `good_renorm.tex`, `shiftcharge_oneloop.tex`, `audit_resonant_channels.tex`, `audit_charges_action.tex`, `audit_projected_measure.tex`, `audit_det_projected.tex`, and the controlling errata. The new scripts and this report use the prefix `shiftcharge`. Existing files and git state were left untouched.

References

[1] K. Hikami, *Baxter Equation for Quantum Discrete Boussinesq Equation*, Nucl. Phys. B **604** (2001) 580–602, arXiv:nlin/0102021.

[2] V. V. Bazhanov, A. N. Hibberd and S. M. Khoroshkin, *Integrable structure of W_3 conformal field theory, quantum Boussinesq theory and boundary affine Toda theory*, Nucl. Phys. B **622** (2002) 475–547, arXiv:hep-th/0105177.

[3] A. A. Belov and K. D. Chaltikian, *Lattice analogues of W-algebras and Classical Integrable Equations*, arXiv:hep-th/9303166.

[4] B. Doyon, J. Durnin and T. Yoshimura, *The Space of Integrable Systems from Generalised $T\overline{T}$-Deformations*, SciPost Phys. **13** (2022) 072, arXiv:2105.03326, doi:10.21468/SciPostPhys.13.3.072.

[5] J. Cardy and B. Doyon, *$T\overline{T}$ deformations and the width of fundamental particles*, JHEP **04** (2022) 136, arXiv:2010.15733, doi:10.1007/JHEP04(2022)136.